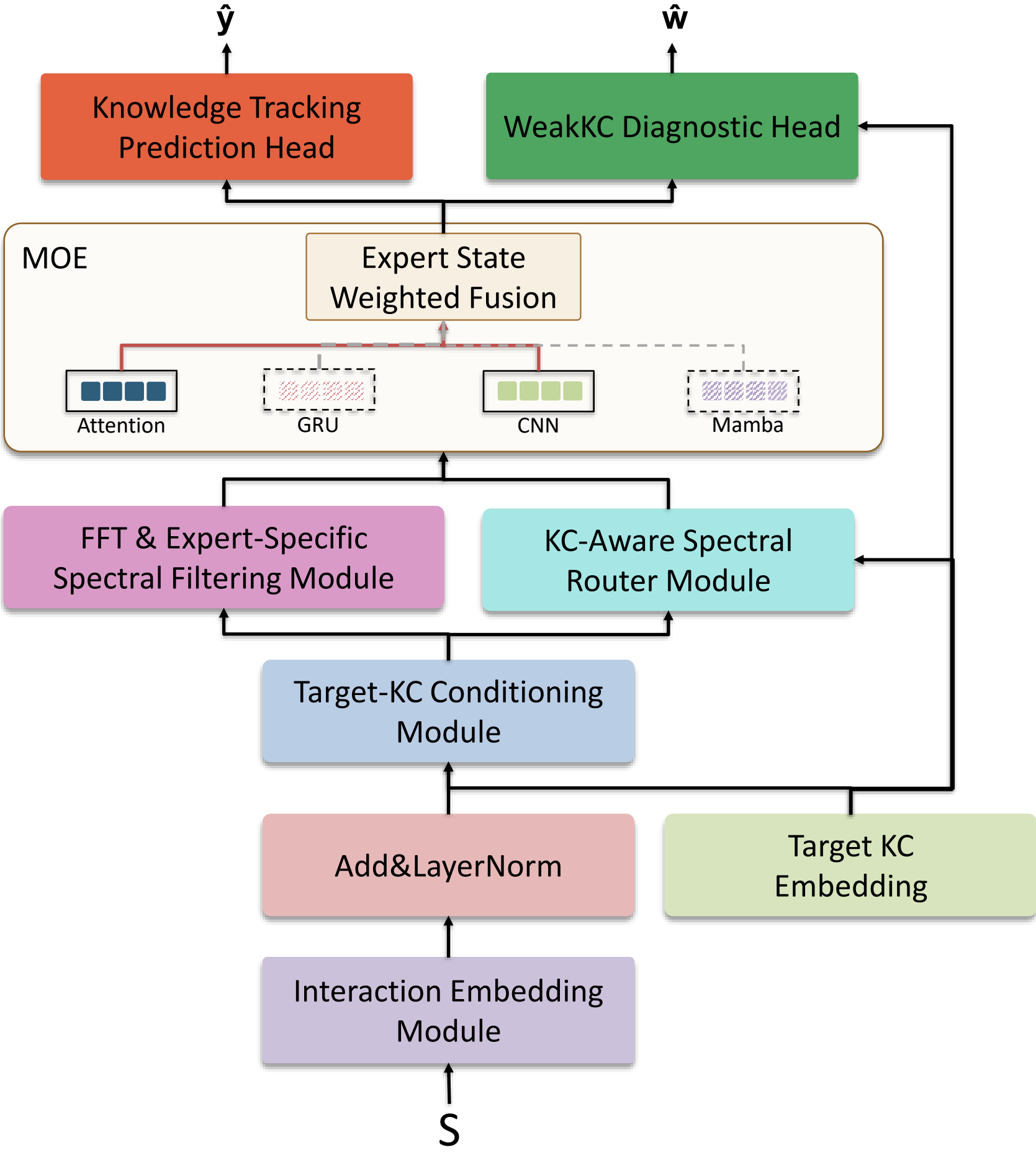
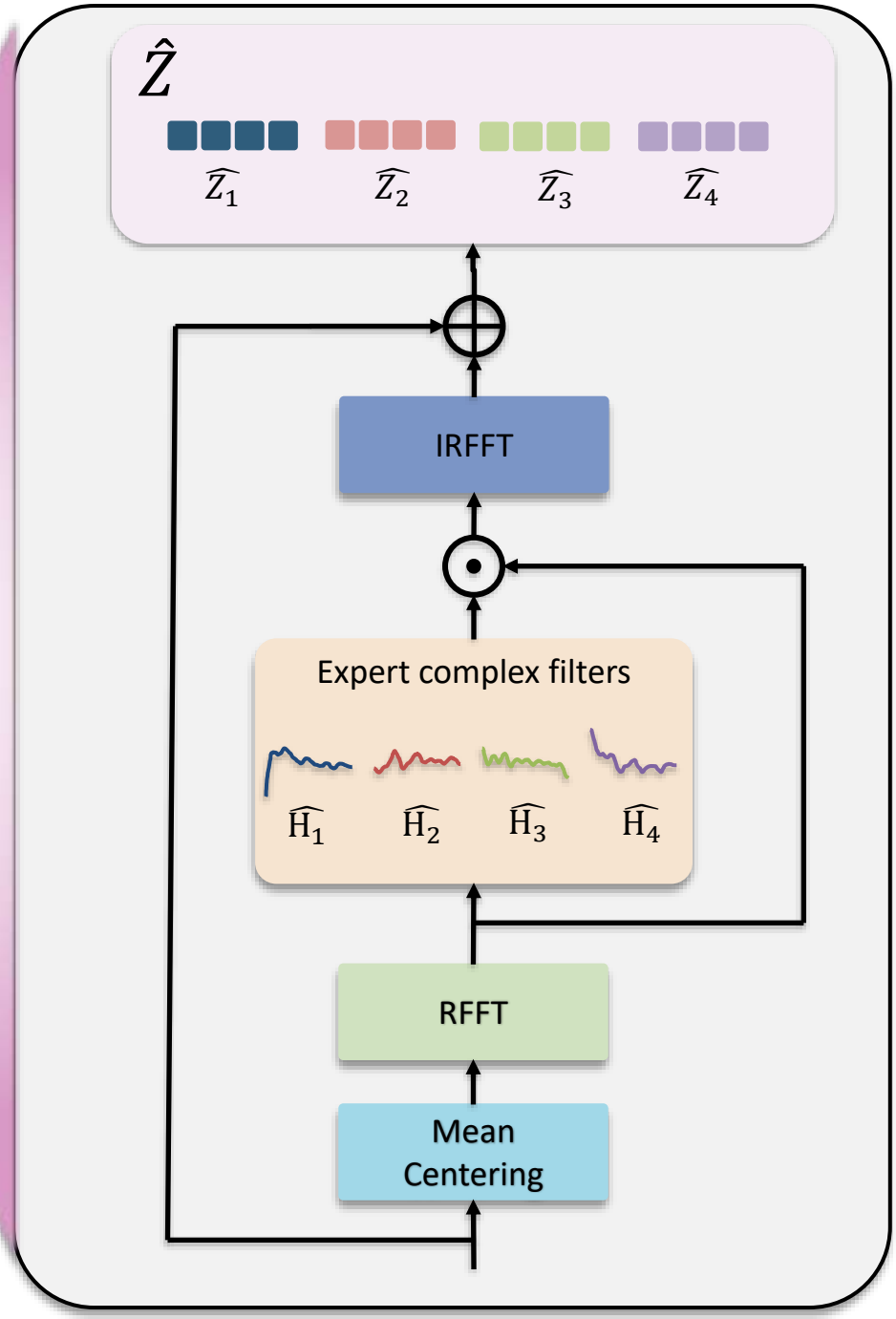
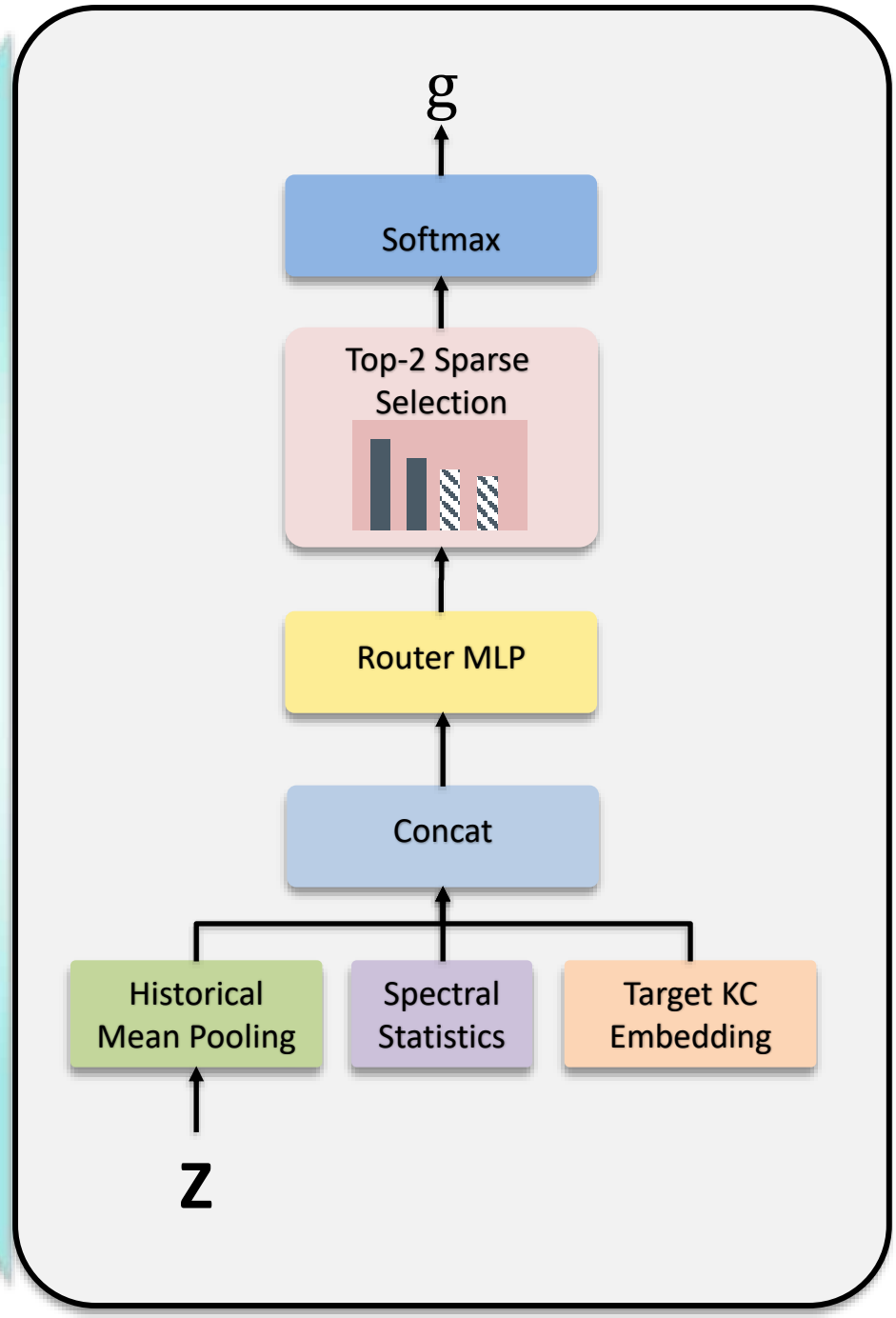
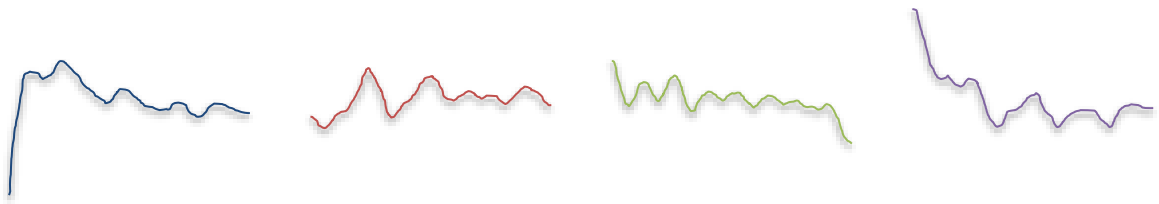

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Knowledge Tracking Prediction Head
WeakKC Diagnostic Head
MOE
Expert State Weighted Fusion
Attention
GRU
CNN
Mamba
FFT & Expert-Specific Spectral Filtering Module
KC-Aware Spectral Router Module
Target-KC Conditioning Module
Add&LayerNorm
Target KC Embedding
Interaction Embedding Module
S

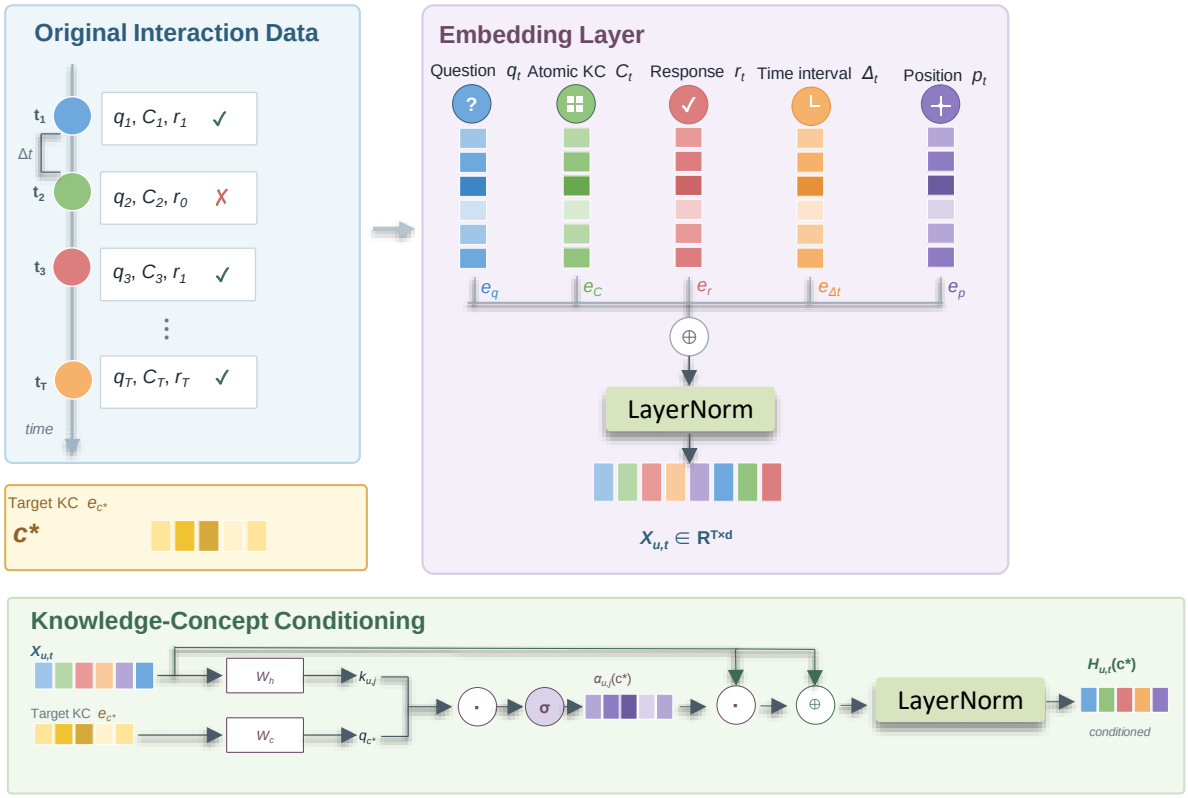
FFT & Expert-Specific Spectral Filtering Module



KC-Aware Spectral Router Module






Original Interaction Data
t1
q1, C1, r1 ✓
Δt
t2
q2, C2, r0 ✗
t3
q3, C3, r1 ✓
tT
qT, CT, rT ✓
time
Embedding Layer
Question qt
Atomic KC Ct
Response rt
Time interval Δt
Position pt
eq
eC
er
eΔt
ep
LayerNorm
Xu,t ∈ RT×d
Target KC ec*
c*

Knowledge-Concept Conditioning
Xu,t
Wh
ku,j
Target KC ec*
Wc
qc*
σ
αu,j(c*)
LayerNorm
Hu,t(c*)
conditioned